

Extracellular matrices regulate extravasation journey of leukocytes and inflammatory tissue fate

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eLife Assessment

This Review Article takes an original angle and covers several aspects of the leukocytes extravasation process with a focus on the role of ECM proteins. It is a timely piece with an original viewpoint. The current manuscript would benefit from improvement in writing and organization.

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Abstract

Leukocyte extravasation across the blood endothelium to inflamed tissues is a crucial defence mechanism against invading pathogens. After elimination of pathogen in the tissue, inflammation needs to be resolved back to steady-state. This cascade comprises of several stages: transmigration through the endothelium and the underlying basement membrane, intra-tissue leukocyte activity, and tissue resolution. In each stage, extracellular matrix proteins in the vascular basement membrane and in tissues regulate a multitude of endothelial and leukocyte functions essential to completion of the cascade, either as a collective force-permissive structure or through signalling by individual matrix proteins. This review will focus on how these extracellular matrices regulate the extravasation journey of leukocytes to illustrate their tight functional inter-dependence with profound impacts on the ultimate post-inflammation tissue fate.

1. Introduction

Cells in an inflamed tissue secrete cytokines to activate local blood vessels and command leukocyte infiltration. Tissue entry of all kinds of leukocytes must be optimally controlled such that the cause of the inflammation, potentially an invading and readily proliferative pathogen, could be timely eliminated. Latency of tissue entry depends on the leukocyte types, typically with neutrophils as the quickest responders (in hours), followed by monocytes (in hours to days) and lymphocytes (in days). Despite the different latency, they often follow common principles for tissue entry. Since the immune attack often involves cytotoxic substances such as granzymes and

proteases, an overdose of leukocyte entry would cause an acute shock and tissue damages. Instead of a passive barrier for leukocytes to penetrate, the endothelium actively controls the leukocyte traffic via expression modulation of apical adhesion or surrogate molecules and junctional remodelling.^{1,2} Only specific sites of the blood vessels, with moderate to low shearing blood flow and optimal expression of supportive endothelial molecules, are permissive to leukocyte exit from the circulation (extravasation). These extravasation sites exist primarily in post capillary venules, venules or specialised blood vessels such as the high endothelial venules (HEV) and the liver sinusoids.²⁻⁴ Understanding regulation details of these extravasation events is fundamental to devise treatments for various diseases.

Along the extravasation journey to fight pathogens, leukocytes are prone to functional modulation, solely through interaction with their surroundings. The resultant leukocyte activities determine whether pathogens can be cleared, and if so, whether the inflamed tissue can be restored to homeostasis. Macrophages derived from either tissue residence or extravasated monocytes play a pivotal role in this latter resolution process.

Across this multi-stage paradigm, extracellular matrix (ECM) proteins in the vascular basement membrane and in tissue play significant regulatory roles. Overviews of the leukocyte extravasation processes, and intra-tissue leukocyte activities regulated by these ECM are given in **Figure 1** and **2** respectively. This review focuses on the regulatory roles of ECM along the extravasation journey of leukocytes to inflamed tissue. This journey has intimate involvement with endothelial cells (EC), especially with their junctional remodelling events. The processes regulated by ECM discussed in this review is summarised in **Table 1**. Following chronological order of the extravasation journey, we will describe what vascular structures constitute extravasation barriers for leukocytes (Section 2–3), how the leukocytes overcome these barriers (Section 4–6), how crossing these barriers affects leukocyte functions (Section 7), and how ECM interaction with leukocyte activities in tissue determines post-inflammatory tissue state (Section 8–10). At the end of this review, we will also discuss why control of macrophage cell state may hold the key for effective modulation of inflamed tissue state to revive tissue homeostasis (Section 10–11).

2. Venular architecture and barriers for leukocyte extravasation

In a blood venule, cells and extracellular components are organized in a defined manner to form extravasation barriers for leukocytes. The most inner circumference of the venule lines a monolayer of EC directly exposed to the shearing blood flow. Except for fenestrated blood vessels, this monolayer is typically continual, with individual EC intercalated via adherens junctions localized at the intercellular basolateral side of the plasma membrane. The endothelial adherens junctions contain various adhesion molecules, such as vascular endothelial cadherin (VE-cadherin), junctional adhesion molecules (JAMs), PECAM-1 and CD99.² In some tissues such as the blood-brain barrier in brain, tight junction molecules such as claudin-5 and ZO-1 are additionally involved. These junctional proteins are not rigidly held in place but under constant balancing actions of endocytosis and exocytosis of junctional proteins such that dynamic remodelling of the junctional tightness is possible. This in turn affects leukocyte traffics and vascular permeability. The most abluminal side of the venule is decorated with scattered perivascular cells (pericytes). (**Figure 1A**) Lacking a definitive pan-pericyte marker, pericytes are defined as mesenchymal cells at vascular proximity, with subsets expressing markers of PDGFR β , CD146, NG2, and/or α -SMA.⁵

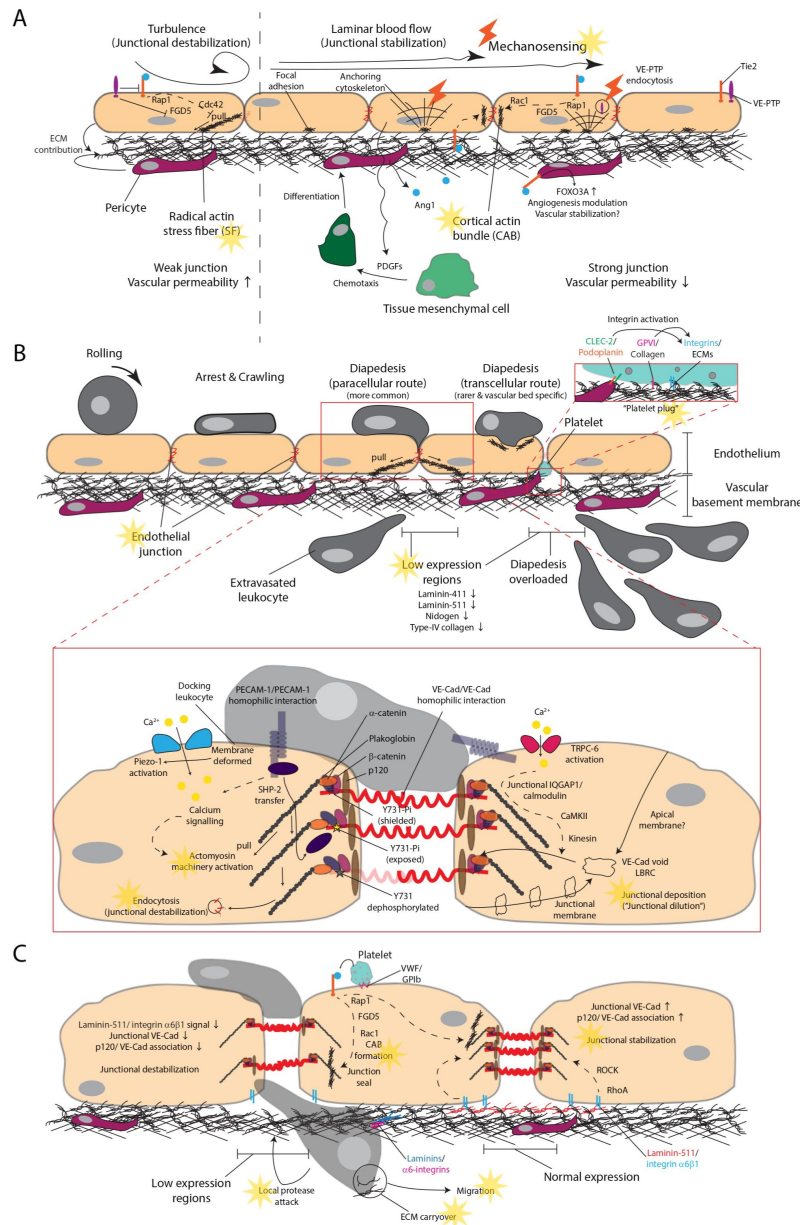


Figure 1

Overview of endothelial functions regulated by vascular basement membrane for allowing leukocyte extravasation.

(A) Vascular basement membrane supports endothelial mechanosensing via focal adhesion to construct leukocyte extravasation barriers by regulating junctional strength and recruiting pericytes. Tie2 signalling is activated by laminar blood flow to trigger force dependent actin filament remodelling to strengthen junctions. (B) Leukocytes prefer to paracellularly transmigrate through the vascular junctional barrier near the low expression regions on the vascular basement membrane. (lower inset) After leukocyte signals diapedesis initiation via calcium signalling, the basement membrane allows force generation in endothelial cells to locally remodel VE-Cadherin junctional barrier for leukocyte passage. (upper inset) Over-usage of low expression regions exposes the basement membrane for partial platelet activation generating a plug to prevent extravasation associated vascular leak. (C) Low expression region loosens endothelial junction and vascular barriers to ease leukocytes passage. At the same time, platelet-endothelium interaction keeps junction surrounding the leukocyte tight to prevent leak. Gold spikes, processes regulated by ECM proteins from vascular basement membrane. Dashed line, multi-step signalling processes.

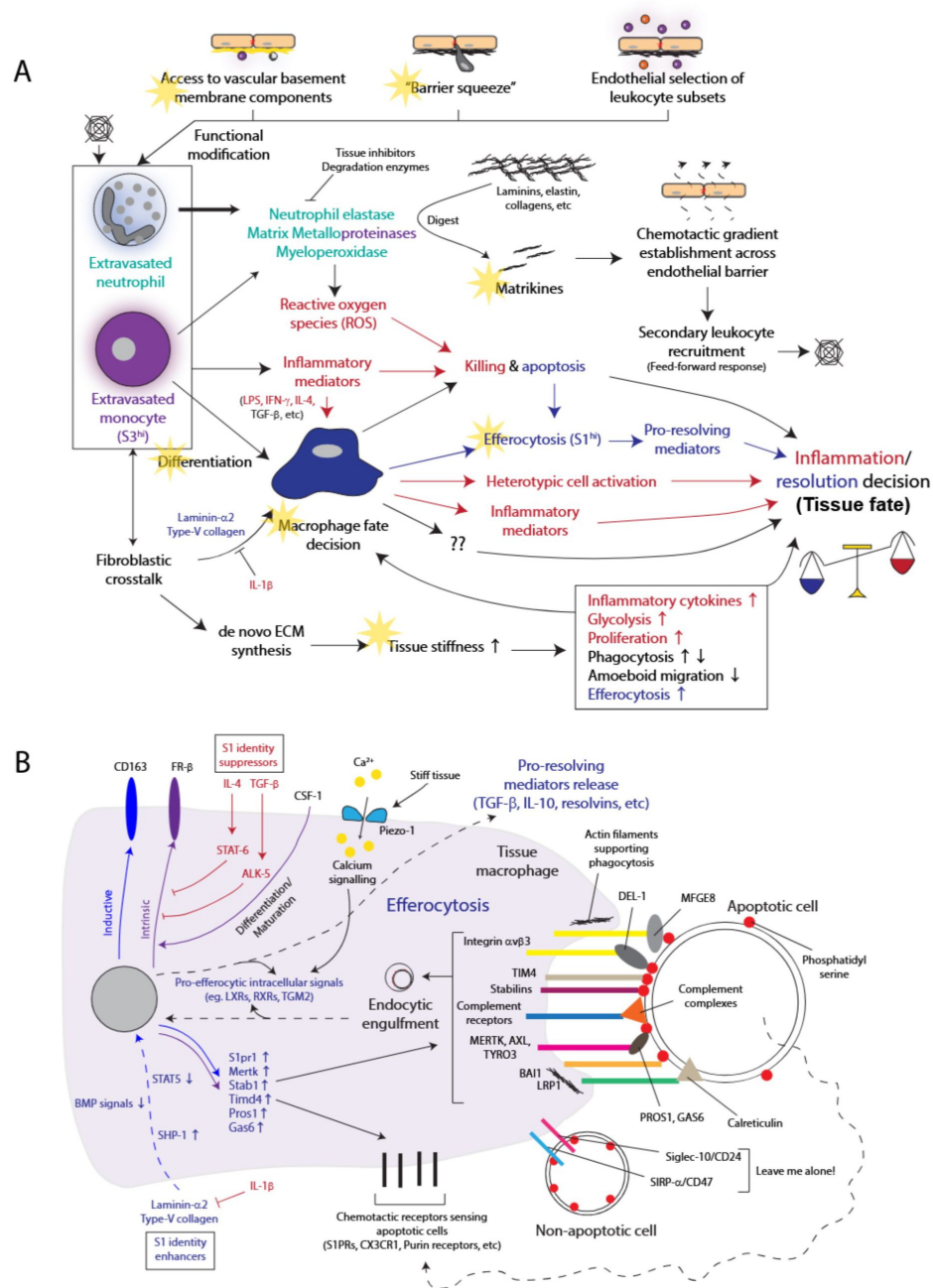


Figure 2

Extracellular matrices modulate extravasated leukocyte activities in tissue leading to fate decision of the inflamed tissue.

(A) After extravasation, neutrophils and monocytes receive constant functional and differentiation influence via interacting with the extravasation barriers, tissue cell and matrix components. Integrating these environmental signals, the consequent leukocyte activities decide tissue fate between sustained inflammation and resolution. (B) Macrophage senses chemotactic materials released by apoptotic cells via an array of receptors to migrate towards and engulf them. Engulfment is mediated by receptor recognition of phosphatidylserine exposed on apoptotic cells either directly or through sandwiching adaptors. Non-apoptotic cells avoid engulfment via specific receptor complexes. Intracellular processing of endocytosed apoptotic cells activates efferocytic programs and releases anti inflammatory mediators. Efferocytic macrophage shows strong S1 identity and the efferocytic capacity is under regulation by external signalling crosstalk. Gold spikes, processes regulated by ECM proteins from vascular basement membrane or in tissue.

Extravasation Journey Stage: Paracellular Trans-Endothelial Diapedesis		
ECM Interaction (Protein or Structure)	Vascular Basement Membrane / Endothelial focal adhesions	
Function		Reference
Sense shear stress in blood flow to secrete PDGFs for pericyte recruitment and Ang1/Tie2-mediated junctional strengthening.		12–14
Sense shear stress in blood flow to endocytose VE-PTP to activate Tie2-mediated junctional strengthening.		15
Enable endothelial force generation pulling on VE-Cad/ catenins/ actomyosin complex to expose Y731 site for dephosphorylation by SHP-2, allowing subsequent VE-Cad endocytosis and local junctional destabilization around the passing leukocyte.		27,28
Enable kinesin mediated delivery of LBRC along an anchored microtubule network to endothelial junction around the passing leukocyte for junctional destabilization.		33–35
ECM Interaction (Protein or Structure)	Laminin-511 / Endothelial Integrin $\alpha 6\beta 1$	
Function		Reference
Activate RhoA/ROCK to promote junctional deposition of VE-Cad to stabilize junction.		38
Promote p120 association to stabilize junctional VE-Cad.		39
Extravasation Journey Stage: Trans-Basement Membrane Diapedesis		
ECM Interaction (Protein or Structure)	Laminins / $\alpha 6$ Integrins	
Function		Reference
Allow migration through the basement membrane barrier.		43,44
ECM Interaction (Protein or Structure)	Vascular Basement Membrane / Leukocyte proteases	
Function		Reference

Table 1

Biological processes regulated by extracellular matrix proteins or structures at each stage along the leukocyte extravasation journey.

Locally breach the basement membrane barrier.		40,44,47
Produce ECM carryovers on extravasated leukocytes.		40,47
ECM Interaction (Protein or Structure)	Laminin carryovers on neutrophils	
Function		Reference
(Unknown)		40,47
ECM Interaction (Protein or Structure)	Lumican carryovers on neutrophils bound by Mac1	
Function		Reference
Promote neutrophil migration via inducing outside-in signalling.		65
ECM Interaction (Protein or Structure)	Vascular Basement Membrane / Neutrophils	
Function		Reference
Deform passing neutrophils and activate Piezo1 to promote bactericidal activity in tissue.		64
Extravasation Journey Stage: Immediately After Extravasation		
ECM Interaction (Protein or Structure)	Collagen / Platelet GPVI	
Function		Reference
Partially activate platelets to adhere on and seal the exposed basement membrane at diapedesis sites with traffics overload.		50,51
ECM Interaction (Protein or Structure)	Vascular Basement Membrane / Endothelial focal adhesions	
Function		Reference
Enable endothelial force generation stimulated by platelet Ang1-Tie2 activation to build junctional stabilizing cortical actin bundles surrounding a passing leukocyte.		55,59
Extravasation Journey Stage: In Tissue		
ECM Interaction (Protein or Structure)	Laminin-511 / Monocyte Integrin $\alpha 6\beta 1$	
Function		Reference

Table 1 (continued)

Table 1 (continued)

Promote monocyte differentiation to macrophage.		66
ECM Interaction (Protein or Structure)	Collagen / Neutrophil MMP-8/9 and Prolyl Endopeptidase	
Function		Reference
Generate the matrikine PGP to secondarily chemoattract neutrophils.		75–78
ECM Interaction (Protein or Structure)	Laminin- γ 2 / Neutrophil Elastase	
Function		Reference
Generate the matrikine FGPPNCEHGAFCSPACYNQVKI to secondarily chemoattract neutrophils.		80
ECM Interaction (Protein or Structure)	Elastin / Macrophage MMP-12	
Function		Reference
Generate the matrikine VGVAPG to secondarily chemoattract monocytes.		81,82
ECM Interaction (Protein or Structure)	Laminin-511 / Leukocyte Proteases	
Function		Reference
Potentially generate fragments containing the matrikine sequence AQARSAASKVKVSMKF to secondarily chemoattract neutrophils and monocytes.		83
ECM Interaction (Protein or Structure)	Stiff Tissue due to ECM Build-up / Leukocyte Mechanosensors	
Function		Reference
Increase inflammatory cytokine production.		89,90,92
Shift leukocyte migration from amoeboid to podosomal mode.		91
Modulate phagocytosis.		91,93,94
Promote phagocyte proliferation.		92
Promote phagocyte glycolysis.		92
Sensitize phagocyte activation response to stimuli.		92
Promote efferocytosis by Piezo1 activation.		103
ECM Interaction (Protein or Structure)	α 2 Laminins and Type-V collagen / (Unknown Monocyte Receptor)	
Function		Reference
Guide monocyte differentiation to efferocytic S1 macrophage by stimulating SHP-1 to suppress STAT-5 activity.		100

In between the EC monolayer and the porous pericyte layer is an ultra-thin adhesive layer of basement membrane composed of a network of self-assembled ECM proteins.^{6,7} Constituent ECM proteins mainly comprise of laminins ($\alpha 4$ and $\alpha 5$ chains), type-IV collagen, nidogens and perlecan.^{8,9} These components bind endothelial integrins organised in focal adhesions to allow EC adhesion,^{9,10} and support establishment of cell polarity and intercellular junctions to limit undesirable tissue access of cellular or soluble blood components. The EC cytoskeleton links to the focal adhesion-ECM structure to form a mechanosensory structure containing proteins (such as talin and vinculin) capable to translate mechanical force into biochemical signals.¹¹ This mechanosensing capacity, at least in part, organises pericytes to perivascular space and subcellular junctional architecture of EC. Laminar shear stress in the blood flow stimulates endothelial abluminal secretion of PDGF-B to recruit tissue mesenchymal cells and differentiate them to pericytes.^{12–14} These pericytes then release Ang1 to activate endothelial Tie2 signalling strengthening endothelial junctions. This junctional strengthening signal is further potentiated by endocytosis of the VE-PTP, which inhibits Tie2, by laminar shear stress.¹⁵ Since both EC and pericytes contribute to ECM components in the basement membrane, this pericyte recruitment process may help formation of nascent basement membrane in angiogenic vessels. (Figure 1A)

For a leukocyte to enter inflamed tissues, it engages in a series of steps well-described by the classical extravasation cascade.^{1,2,16} After firm adhesion on the endothelium, down-tuning of integrin activity allows the leukocyte to crawl on the endothelium and search for a suitable tissue entry site.^{17,18} At this stage, the continual endothelial monolayer and the underlying basement membrane are the sole barriers for leukocytes to overcome before tissue access. (Figure 1B) The porous abluminal pericyte layer, in contrast, does not barricade leukocyte access. These barrier crossing processes, known as diapedesis, need to be tightly controlled to permit extravasation without disrupting general vascular structural integrity.

3. Paracellular versus transcellular route of entry across the endothelial barrier

A leukocyte could transmigrate across the endothelium either by diapedesis through the junctions (paracellular route) or directly through the EC (transcellular route) *in vitro*.^{19–21} This observation questions whether the endothelial junction or the apical surface represents the major endothelial barrier for leukocytes under physiological conditions. If one route can be specifically blocked, assessment of its influence on the extravasation response shall inform the importance of the blocked route.

VE-cadherin is a major regulator of the adherens junction and supports junctional integrity by homophilic dimerization of VE-cadherin from adjacent cells. Like other cadherins, VE-cadherin is bound at the C-terminus by intracellular β -catenin and α -catenin, which is then connected to the actomyosin contraction machinery. (Figure 1B, lower inset) Since association to α -catenin stabilizes cadherin adhesion,²² association or dissociation of catenin from VE-cadherin allows fine-tuning of junctional stability. In a previous study, transgenic mice with VE-cadherin genetically fused to α -catenin, thereby stabilizing the VE-cadherin junction and limiting paracellular diapedesis, were generated.²³ In these mice, the paracellular diapedesis events were reduced by ~50%, but the transcellular events were unaffected. In IL-1 β inflamed cremaster, LPS stimulated lung, and a skin model of delayed type hypersensitivity, both neutrophil and lymphocyte extravasation reduced by ~60%.²³ The similar reduction magnitudes of extravasation and paracellular diapedesis events suggest most of the reduced extravasation was due to blockade of paracellular diapedesis.

Although these observations suggest the paracellular endothelial junction is the sole barrier a leukocyte needs to overcome in vivo, “closing” the paracellular route did not affect lymphocyte entry to lymph nodes via the HEV, indicating vessel- or tissue-specific preference of diapedesis route.²³ Later, it was demonstrated that the density of ICAM-1, an adhesion molecule supporting leukocyte adhesion on the endothelium, presented by the EC could influence the route decision, that a high ICAM-1 density favours the transcellular route for helper T cells across brain EC.²⁴ Hence, the preference for transcellular or paracellular route appears to depend on the adhesion strength between the leukocyte and the endothelium and the vascular bed in concern. Nevertheless, in many inflammatory conditions, the paracellular route of transendothelial migration remains the major route.²³ The intercellular adherens junctions therefore are the major barriers for leukocyte to cross the endothelium. (**Figure 1B**)

4. Vascular basement membrane supports force-dependent endothelial junctional remodelling required for leukocyte diapedesis across the endothelium

Actomyosin contraction in EC is required to support leukocyte paracellular transmigration.^{25,26} The stable EC anchorage on the basement membrane via focal adhesions allows contractive force to be transmitted to actomyosin filaments at the cell periphery. It was hypothesized that such force pulling on endothelial junctional proteins helps open the junction for leukocyte passage, but molecular details had been obscure until recently.

Using a Förster resonance energy transfer (FRET) based molecular force sensor module inserted to the C-terminus of VE-cadherin, which the FRET signal vanishes when the pulling force exceeds 4 pN, it has been elegantly demonstrated that endothelial mechanical force acting on VE-cadherin triggers a cascade of events that allow paracellular passage of a leukocyte.²⁷ Multiple tyrosine residues, which phosphorylation states control different endothelial functions, are present in VE-cadherin. Shown in vivo with knock-in point mutations, dephosphorylation of tyrosine at Y731 position is required for leukocyte extravasation to inflamed tissues.²⁸ Upon leukocyte docking, EC-leukocyte PECAM-1 homophilic interaction triggers calcium signalling to activate the actomyosin machinery that pulls on VE-cadherin. The pulling force exposes the catenin-shielded Y731 site for dephosphorylation by the phosphatase SHP2, and induces endocytosis of VE-cadherin to loosen the junction for leukocyte passage.²⁷ (**Figure 1B, lower inset**) Since the force magnitude a cell generates correlates to the substrate rigidity,²⁹ endothelial cells on a soft substrate cannot dephosphorylate Y731 showing force dependence of this process. Fluorescence lifetime imaging microscopy (FLIM) is a technique that measures the lifetime of a fluorophore, defined as the time the fluorophore stays excited, which depends on the fluorophore species and its surrounding microenvironment.³⁰ Alteration in the FRET signal, due to the pulled VE-cadherin tension sensor, incurs a shift in fluorescence lifetime detectable by FLIM. FLIM images showed that only the local junction surrounding the passaging leukocyte was being pulled to not to compromise the overall junctional integrity.²⁷ Since not all junctions within the same EC were pulled, there involves a subcellular spatial restriction to confine the initiating calcium signal near the junction being pulled.

Such a subcellular cue might be provided by the leukocyte locally pressing the endothelial plasma membrane. A recent report describes the involvement of the tension sensitive cation channel Piezo-1 in transendothelial diapedesis. Leukocyte docking changes the tension in plasma membrane and provokes calcium influx via nearby Piezo-1 to activate the contractile actomyosin machinery.³¹ (**Figure 1B, lower inset**) This model implies localized calcium signal and thus

the subsequent actomyosin contraction only occur close to the leukocyte dock site. One would still need to explain why the leukocyte arrest site is not always the diapedesis site that a leukocyte usually crawls over a distance before beginning diapedesis.^{18,32}

There are other endothelial force dependent processes supporting trans-endothelial diapedesis. Studies suggested directed membranous deposition, termed lateral border recycling compartments (LBRC), to the junction facilitates leukocyte passage. Deposition of LBRC to leukocyte-proximal junction requires force generation by the molecular motor kinesin along anchored microtubules.³³ Mechanistically, LBRC contains all other diapedesis-relevant junctional proteins but not VE-cadherin, and deposition to junctional site surrounding a transmigrating leukocyte dilutes the local VE-cadherin density and loosen the junction.³⁴ Regarding how LBRC spatially coordinates with the passing leukocyte so that deposition is only directed to the leukocyte-contacted junction, again calcium signalling is involved. With calcium sensor mice that fluorescently visualise calcium signalling, leukocyte diapedesis triggers calcium signalling localized to the proximal junction. In this model, EC-leukocyte PECAM-1 homophilic interaction initiates calcium influx through the nearby calcium channel TRPC6. This activates calmodulin, immobilized at the junction via IQGAP1 binding, and downstream CaMKII to drive junctional deposition of LBRC.³⁵ (Figure 1B, lower inset) TRPC6 can be activated by mechanical stress,³⁶ it might as well be activated by direct leukocyte pressing like Piezo-1. While it is known that kinesin-microtubule machinery can be regulated by calcium signals,³⁷ the molecular details on its coupling to LBRC movement are unclear. Additional open questions like whether and why LBRC membrane is exclusively from junctional membrane, and how VE-cadherin is excluded from entering this compartment remains to be explored.

Overall, junction opening during paracellular transendothelial diapedesis involves numerous force dependent processes and the basement membrane anchors endothelial cytoskeleton to allow force generation in EC.

5. Pre-existing barrier weak spots guide leukocyte diapedesis

Organized laminin structures known as the “diapedesis portals” in the venular basement membrane directly signal endothelial junctional remodelling to regulate leukocyte diapedesis. Laminins are heterotrimers of the constituent chains (α -, β and γ -chains). Laminins $\alpha 4 \beta 1 \gamma 1$ and $\alpha 5 \beta 1 \gamma 1$ (laminin-411 and laminin-511) are the major laminins in the vascular basement membrane.⁸ In vitro assays showed that laminin-511, but not laminin-411, activates RhoA/ROCK in EC to trigger junctional translocation of VE-cadherin, which stabilizes the junction and suppresses leukocyte transendothelial migration.³⁸ In the presence of shearing blood flow, laminin-511 also promotes p120 association to VE-cadherin and further stabilizes its presence at the junction.³⁹ (Figure 1C) These junction stabilizing properties of laminin-511 leads to the postulation that, if there are sites on the basement membrane with lower laminin-511 density, endothelial junctions near those sites would be more permissive to leukocyte passage. Such sites have actually been demonstrated.

Careful and detailed examination of the venular basement membrane ultrastructure by confocal microscopy with 3D-reconstruction revealed that ECM component distribution is uneven and patchy with regions of low expression of laminin-511, laminin-411, nidogen and type-IV collagen.^{40,41} The spatial analyses revealed the luminal sides of these low expression regions are close to endothelial junctions whereas the abluminal sides are typically not occupied by pericytes.⁴⁰ (Figure 1B) The reduced laminin-511 density at these regions locally “loosened” the proximal luminal endothelial junctions for easier leukocyte access. Indeed, leukocyte diapedesis events were observed close to these low expression regions.^{40,42} Trans-basement

membrane diapedesis through these sites still requires $\alpha 6$ integrin to allow migration on laminins and protease activity to “drill through” the basement membrane.^{43,44} (Figure 1C) In mice lacking laminin- $\alpha 4$, there is a compensatory upregulation of laminin- $\alpha 5$ resulting in a less patchy expression in the basement membrane. These mice served as knockout analogy of the low expression regions, and reduced leukocyte extravasation was recorded confirming the facilitative role of the low expression regions.^{38,45}

The exact mechanism how these low expression regions are constructed in the first place remains elusive. One possibility is that these low expression regions might be remnants of previous diapedesis events, which involve protease activities. As neutrophils pass through these low expression regions, these regions transiently enlarge with proteases involvement including neutrophil elastase; the sizes recover to basal level over time.^{40,46} The enlargement is at least in part related to deformability of the leukocyte since the passage of the more rigid neutrophils, but not the more flexible monocytes, resulted in low expression region enlargement.⁴⁷ In parallel, gaps between pericytes on the abluminal side of the basement membrane enlarge.⁴⁸ An implication of these observation is that pericytes may have stretched these low expression regions in the basement membrane to enlarge them. Alternatively, the gap enlargement might be due to stretching by the rigid neutrophils or proteolytic activities followed by ECM redeposition. Direct visualization remains to be demonstrated, potentially by live confocal imaging and fluorescently labelled leukocytes, basement membrane and pericytes. Whether and which component(s) of the basement membrane could be labelled by fluorescent proteins without disrupting its native organization remains to be investigated. It is unclear whether de novo deposition or the postulated basement membrane stretching by pericytes contributes to the recovery. If de novo deposition is involved, it remains to investigate how the deposition is controlled as not to bury and destroy these sites.

6. How do leukocytes overcome barriers without vascular leak?

During paracellular transendothelial diapedesis, only the endothelial junction surrounding the passing leukocyte is loosened. When asked if such localized junctional destabilization alone is sufficient to avoid vascular leak, it seems not to be the case. Complementary mechanisms must be involved, or the diapedesed leukocyte would leave a confined but leaky pore behind. This diapedesis pore must be sealed, either by endothelial junctional remodelling immediately behind the passing leukocyte or by a simple plug to prevent leak. This essential seal is provided by platelets. The vascular basement membrane coordinates these platelet processes through transient exposure of collagens or providing a structural support enabling force-dependent endothelial junctional remodelling.

In thrombocytopenic mice, which lack circulating platelets, skin inflammation induced by either UVB, croton oil or immune complex (reverse passive Arthus reaction) caused bleeding at the inflamed site. This bleeding can be reduced by either leukocyte depletion or suppression of leukocyte-endothelial interaction via rolling/ adhesion blockade or chemokine neutralization. Importantly, bleeding spots colocalized with extravasation sites and specific blockade of leukocyte diapedesis with VE-Cadherin Y731F point mutation reduced bleeding. These suggest leukocyte diapedesis in the absence of platelets caused the bleeding.⁴⁹ Similar inflammatory bleeding could be observed in GPVI-/-mice which platelets are present in blood but do not express GPVI.⁵⁰ GPVI is an ITAM-containing receptor that signals via downstream Syk and Btk and activates platelets upon binding collagen, which is present in the vascular basement membrane or in tissue. Transfusion of untreated, but not Syk or Btk inhibited, wildtype platelets to GPVI-/-mice rescued the inflammatory bleeding, indicating GPVI/Syk/Btk signalling is required.⁵⁰ These findings imply platelet contact with the collagenous vascular basement membrane signals and activates

platelets via GPVI to plug the diapedesis pore left by the extravasated neutrophil. This has been recently directly visualized in a skin immune complex vasculitis model with real-time 4D intravital confocal microscopy that single platelets immediately adhered and plugged the blood vessel at the exact diapedesis site utilized by the passing neutrophil. Examination of ultra-structures with transmission electron microscopy found granules in the plugging platelets suggesting these platelets had not been fully activated and degranulated.⁵¹ Consistently, a previous report showed platelet granules are not required for bleeding prevention in this inflammation model.⁵² Simultaneous antibody blockade of both GPVI and CLEC-2 resulted in much more bleeding, as well as more platelet leakage to extravascular space beyond the basement membrane, than GPVI single blockade.⁵¹ CLEC-2 binds podoplanin that is present on pericytes but normally not on EC nor in the basement membrane. Since both GPVI and CLEC-2 are activating receptors and platelet adhesion on the underlying basement membrane ECM likely involves integrins, GPVI and CLEC-2 probably concert in transmitting inside-out signalling for integrin activation to support adhesive plugging. (**Figure 1B, upper inset**)

There is another mechanism where platelets prevent vascular leak via endothelial junctional remodelling. Inflamed EC display von Willebrand factor (VWF) on the apical surface allowing platelet interaction.^{53,54} VWF binding via GPIb activates platelets to lumenally release angiopoietin-1 which activates Tie-2.^{55–57} Activated Tie-2 phosphorylates downstream Rap1 and FGD5 which serves as a guanine exchange factor for Rac1 and Cdc42 to suppress stress fibres (SF) and to promote circumferential actin bundles (CAB).^{58,59} With a stable anchorage on the basement membrane such that tension of actin filaments could be exerted on junction, CAB strengthens whereas SF loosens endothelial junctions. (**Figure 1C**) Endothelial conditional knockout or siRNA knockdown of Tie-2 phenocopied neutrophil diapedesis induced bleeding observed in thrombocytopenia mice.⁵⁵ How this junction sealing process coordinates with leukocyte passage is unclear, or it just constantly seals any junctional pore incurred by leukocyte diapedesis.

If this Tie-2 centred mechanism is in place, why is platelet plugging necessary in immune complex-mediated inflammation to prevent diapedesis mediated bleeding? It was found that platelet plugging occurred not at random diapedesis site but was concentrated at hotspots experiencing on average 4 successive neutrophil diapedesis events.⁵¹ It thus appears that this Tie-2 mechanism is the default measure to seal the diapedesis pore, but may not catch up to the pace of leukocyte diapedesis in some inflammatory scenarios. On the other hand, leukocytes tend to take the partially breached, easier route for diapedesis. When the diapedesis breach overwhelms the Tie-2 mechanism to an extent that exposes the collagenous basement membrane, the platelet plug serves as the final defence line to prevent bleeding. (**Figure 1B**)

Both Tie-2 and the downstream FGD5 are substrates of the inhibitory phosphatase vascular endothelial protein tyrosine phosphatase (VE-PTP). The Tie-2 mechanism could likely be strengthened by suppressing VE-PTP activity to counter vascular destabilization in inflammatory conditions with too rapid leukocyte extravasation. This approach has been shown to be beneficial in other vascular pathologies (reviewed in ⁶⁰). Alternatively, there have been proposals to use platelet as a drug delivery vehicle to tumours exploiting the thrombotic leaky tumoral vasculature.^{61,62} As both luminal VWF binding and the platelet plug involve partial platelet activation, similar platelet engineering approaches may be attempted to deliver inflammatory modulators in inflammatory diseases.

7. Barrier passages modify leukocyte functions in tissue

When a leukocyte diapedes through the endothelial barrier, the cell body is heavily deformed with endothelial junctional proteins constantly tightening up the endothelial-leukocyte cell surface interface to prevent leakage. Similar deformation is recorded when a leukocyte passages through the basement membrane barrier with the uses of barrier-degrading proteases. These are biophysically and biochemically intense processes that could have changed the functional phenotype of the extravasated leukocyte. In addition, unique cellular interaction with EC or material interaction with the otherwise shielded basement membrane during extravasation may also modify leukocyte functions. Growing evidence have supported this view.

It has been recently shown that in neutrophils, “squeezing” through confined space, corresponding to either the endothelial or basement membrane barrier in vivo, activates the tension sensor Piezo-1 to trigger calcium signalling and to upregulate expression of the cytotoxic hydrogen peroxide producing enzyme NOX4⁶³, resulting in increased bactericidal activity.⁶⁴ Proteolytic activities involved during trans basement membrane diapedesis of neutrophils resulted in carry-over of laminin bits on cell surface after arriving at tissues.^{40,47} While these bound laminins are bioactive as aforementioned, it is unclear whether they modulate the neutrophil functions or interfere with crosstalk to other cell types. A potential hint comes from a similar carry-over event with lumican, which is expressed by endothelial cells but not neutrophils. Lumican fragments were found on extravasated peritoneal neutrophils, but not on pre-extravasated blood or bone marrow neutrophils, and supported chemotactic migration and intra-tissue agility.⁶⁵ The enhanced motility likely involves lumican/Mac-1 mediated outside-in signalling. (Figure 1C)

For monocytes, any functional influence by extravasation may occur immediately at the level of monocyte, during macrophage differentiation, or at the level of differentiated macrophage. A previous study showed that following trans-basement membrane diapedesis, some extravasated monocytes remained associated with the basement membrane where laminin-511 promotes transition of Ly6C⁺MHC-II^{lo} monocytes to Ly6C⁺MHC-II^{hi} macrophages.⁶⁶ While this function is consistent to the perivascular macrophages abundantly seen, it is unclear whether this differentiation boost is possible because passaging monocytes are in contact with the basement membrane in the first place, or monocytes have a natural tendency to migrate towards blood vessels. The latter scenario would argue this basement membrane function is standalone and does not require the extravasation process, that basement membrane laminin-511 would support monocyte differentiation even if monocytes are artificially injected into tissue. Either way, physiologically, exposure to laminin-511 in the basement membrane is only possible after extravasation.

Besides, monocyte/macrophage functions are affected by ontogeny (embryonic progenitor derived versus bone marrow derived) and lineage origins.^{67,68} Previously described with inducible lineage tracing mice, the hematopoietic stem cells in bone marrow are partially derived from PDGFRα⁺ mesenchymal progenitor transiently present during embryonic day E7.5-E8.5 (PDGFRα-lineage).⁶⁹ As a result, about one-fourth of circulating monocytes is from the PDGFRα-lineage. Interestingly, vasculatures in skin and colon selectively prefer extravasation of PDGFRα-lineage Ly6C⁺ monocytes. The responsible molecules nevertheless remain unclear. After differentiation to macrophages, PDGFRα-lineage cells show reduced DC-like characters and express a lower level of level of the inflammatory *Il1b* transcripts, suggesting less involvement in activating cellular engagement.⁷⁰

Collectively, functions of neutrophils and monocytes are modulated either directly via signalling events incurred by the extravasation processes, via selective extravasation of a leukocyte subset, or via novel access to function-modulating materials in tissues. (Figure 2A [↗](#))

8. Leukocyte proteases fragment tissue ECM to promote secondary leukocyte trafficking

Proteases are powerful weapons utilized by innate leukocytes, especially neutrophils, to fight inflammation triggering invading pathogens after tissue entry. Extravasated neutrophils have experienced the “squeeze” by the endothelial and basement membrane barriers and hence exhibit even stronger aggression towards pathogens. Delivery of neutrophil proteases stored in intracellular granules could take several forms. Besides the typical degranulation, neutrophils may choose to release neutrophil extracellular traps (NET) which comprise of histone-bound chromatin, proteases, and additional offensive enzymes such as the reactive oxygen species (ROS) generating myeloperoxidase.⁷¹ [↗](#) While a neutrophil cannot survive chromatin loss, in exchange, the sticky NET efficiently entangles pathogens for concentrated protease attack. Alternatively, proteases may be released as exosomes where individual molecules are oriented to resist antiprotease actions and to achieve stronger activities than free proteases.⁷² [↗](#),⁷³ [↗](#) However, protease activities not only destroy pathogens but also tissue ECM. Excessive protease activities could damage tissue integrity, as observed in chronic inflammation such as cystic fibrosis and chronic obstructive pulmonary disease.⁷³ [↗](#),⁷⁴ [↗](#)

Breakdown of tissue ECM produces a class of biologically active ECM degradation products termed matrikines, that regulate leukocyte trafficking. These matrikines are often small enough to permeate the endothelial barrier to form a functional chemotactic gradient in vivo. Interestingly, a matrikine tends to promote chemotaxis of the leukocyte type releasing its synthesizing protease(s), constituting a feed forward positive feedback loop. (Figure 2A [↗](#))

Proline-Glycine-Proline (PGP) is a tripeptide derived from collagen under actions of neutrophil protease MMP-8/9 and prolyl endopeptidase.⁷⁵ [↗](#) Acetylation at N-terminus of PGP enables chemotactic activity for neutrophils via binding chemokine receptor CXCR1/2.⁷⁶ [↗](#)–⁷⁸ [↗](#) This binding is explained by the structural similarity between PGP/CXCR2 and the binding of the cognate ligand KC to CXCR2.⁷⁸ [↗](#) Leukotriene A4 hydrolase (LTA4H), which generates the well-known strongly chemotactic leukotriene B4, serves a surprising function to degrade PGP.⁷⁹ [↗](#) Another peptide corresponding to the 597–618 region of human laminin-γ2 chain was found to be chemotactic for neutrophils. Release of this peptide requires neutrophil elastase activity.⁸⁰ [↗](#) Proteolytic cleavage of elastin, abundantly found in elastic tissues such as skin and lung, by macrophage MMP-12 produces a hexapeptide VGVAPG to chemoattract monocytes via binding S-Gal receptor.⁸¹ [↗](#),⁸² [↗](#) Laminin-511 in the vascular basement membrane potentially produces matrikines. A synthetic peptide (AQARSAASKVKVSMKF) derived from laminin-511 chemoattracts neutrophils and monocytes in vitro.⁸³ [↗](#) But it is unknown if this fragment is physiologically generated and if bioactivity requires exact sequence and length.

These matrikines are regulated by intra-tissue availability of synthesis and degradation enzymes, as well as their regulators. For example, the presence of low-molecular weight hyaluronan stimulates tissue MMP-9 expression⁸⁴ [↗](#) and may promote formation of PGP. Smoking strengthens macrophage expression of MMP-12 and suppresses the degradative LTA4H activity in lung could respectively promote accumulation of VGVAPG and PGP.⁸⁵ [↗](#),⁸⁶ [↗](#) Bioactivity of MMPs, which are typically secreted as latent proproteins, requires activation. The zinc ion at the catalytic site, which is otherwise shielded by the thiol group of a cysteine residue of the pro-domain, needs to be exposed. Detailed activation mechanisms vary by the MMP species and are not fully known, but could involve cleavage of the pro-domain by furin or an active MMP or allosteric exposure of the

catalytic site.⁸⁷ On the other end, active MMPs could be inactivated by tissue inhibitors of metalloproteinases (TIMPs). Thus, matrikines contribute to secondary leukocyte recruitment caused by protease activities of extravasated leukocytes and further potentiate leukocyte activities in tissue. The multi-level regulation of matrikine ana-/cata-bolism implies their contribution to leukocyte activities in tissue likely vary by diseases; further investigations shall identify common features of conditions where managing matrikine-induced secondary leukocyte trafficking presents benefits against diseases.

9. Crosstalk between extravasated leukocyte activities and tissue stiffness

On the contrary, instead of ECM degradation, there are disease conditions like skin keloid and bleomycin induced pulmonary fibrosis where inflammatory activities of extravasated leukocytes promote amorphous collagenous deposition and harden the tissue. Keloid is considered as a form of pathological scarring where initial wounding does not properly resolve but persists as chronic inflammation with undesirable accumulation of ECM. Without knowing the exact cause, multiple leukocyte types (including macrophages, mast cells and regulatory T cells), growth factors and cytokines that promote fibroblast proliferation (TGF- β , PDGFs, FGFs) and angiogenesis (FGFs, VEGFs), and additional pleiotropic cytokines (IL6, IGF-1) contribute to this condition.⁸⁸ Similarly, following an injury by excessive bleomycin, restless leukocyte activities in lung stimulates overgrowth of ECM-secreting fibroblasts. ECM build-up these pathological scenarios alters tissue stiffness that translates to profound impacts on leukocyte functions. (Figure 2A)

Many leukocytes are mechanosensitive and change their behaviours according to the surrounding stiffness. Analogous to the EC-basement membrane described earlier, leukocytes bind the surrounding ECM substrates via focal adhesion to transmit the mechano-signals. In general, tissue hardening due to ECM accumulation drives inflammatory functions of leukocytes. Monocytes and T cells enhanced their secretion of inflammatory cytokines in a stiff environment.^{89,90} Most leukocytes migrate in amoeboid fashion in normal tissues. A study has shown that macrophages adopted this migration mode only on soft substrates (<88 kPa); on stiffer substrates (323 kPa), cells instead exhibited podosome-dependent migration.⁹¹ Many other leukocyte qualities relevant to inflammation, such as proliferation, glycolytic metabolism and sensitivity to stimulus, are promoted by stiff substrates.⁹² Phagocytic capacity may also be modulated by stiffness but contradicting reports exist.^{91,93} One reason for this is that stiffness regulates the expression of specific phagocytic receptors and could differently affect phagocytosis towards different targets. Macrophage mannose receptor (CD206), which binds and supports endocytosis of ovalbumin, was downregulated on a stiffer substrate (12–50 kPa), when compared to a softer substrate (2 kPa). Accordingly, ovalbumin phagocytosis was reduced on the stiffer substrate, but transferrin endocytosis was unaffected.⁹⁴ Interestingly, this is an effect triggered only by medium stiffness since cells cultured on hard plastic (3 GPa) expressed more CD206. Such a non-linear effect on phagocytosis and the different ranges of substrate stiffness may explain the discrepancy between reports. In vivo studies of stiffness effects (e.g. hydrogel implantation) could be tricky without causing, or at least consideration of, structural disruption of the native tissue environment.

10. Macrophage efferocytosis is critical to timely subside inflammation and restore tissue homeostasis

To avoid these pathological events caused by abnormal and prolonged leukocyte activities, inflammation must be timely tamed as soon as the triggering pathogen is cleared. While pathogen removal reduces inflammatory reaction, many of the reactions triggered such as the matrikine

response and stiffness driven inflammation are self-sustaining. Active processes must be involved to stop inflammatory propagation.

Since neutrophils, which are the first to arrive at the inflamed tissue, have a relatively short lifespan (in days), considerable number of apoptotic neutrophils have been accumulated by the time of pathogen clearance. Somatic cells damaged by inflammatory mediators add to the apoptotic counts. Macrophages are responsible for clearing up these dying cells to prevent necrotic release of intracellular materials that can trigger secondary tissue damages. Macrophages sense the chemotactic materials released by apoptotic cells and migrate towards them. Different from endocytosis of soluble and smaller particles, engulfment of apoptotic cells, either directly via receptor binding or indirectly via bridging adaptors, evokes intracellular signalling programs to acquire a pro-resolving cell state. This process, termed efferocytosis, produces anti-inflammatory mediators, such as IL-10 and resolvins, as end products.⁹⁵ (Figure 2B) Efferocytosis thus serves as a switch for transiting inflammation to resolution. However, acquisition of the efferocytic phenotype is under competition of other fate guiding signals such as pathogen derived LPS, IFN- γ , TGF- β , IL-4 and IL- β , which promotes killing responses, cross-activates other leukocytes like class-II helper T cells, or reduces tissue availability of efferocytic supportive signals.^{96–98} (Figure 2A)

Efferocytosis could be facilitated with soluble factors produced by other cell types. Shown in ligation induced periodontitis and monosodium urate crystal induced peritonitis models, EC produces soluble DEL-1 that bridges phosphatidylserine exposed on apoptotic cells and integrin $\beta 3$ on macrophages to ease recognition of target cells and facilitate efferocytic responses.^{95,99} Recently, fibroblastic ECM has been shown to guide monocyte differentiation to macrophage of strong efferocytic capacity in a vitamin D3 analogue induced atopic dermatitis model. In skin, monocytes mainly extravasate in hypodermis where laminin- $\alpha 2$ isoforms and type-V collagen were observed to colocalize. These ECM activate the phosphatase SHP1 to suppress STAT5 activity during CSF-1 driven differentiation to form FR β /CD163+ S1 macrophage, with high expression of both chemotactic and efferocytic receptors towards apoptotic cells.¹⁰⁰ Macrophage is responsive to BMP signalling,^{101,102} and endogenous BMP signals may also be involved in forming S1 macrophage. Like STAT5, suppressing BMP signals directly induced the S1 phenotype and prevented further induction by ECM. Boosting S1 differentiation with exogenous laminin-211 resolved atopic dermatitis; failure to form sufficient S1 macrophage, on the other hand, induce eosinophil maladaptation tilting the dermatitis away from resolution.¹⁰⁰ Intriguingly, while tissue hardening due to ECM build-up generally supports inflammation, a recent study showed that it also enhances macrophage efferocytosis via Piezo-1 activation, serving as an apparent counter-measure for the stiffness-induced inflammatory responses.¹⁰³ (Figure 2B) Efferocytosis effectors have been proven beneficial in various experimental infections in peritoneal cavity, liver, gastrointestinal tract, lung and even the brain.¹⁰⁴ This process therefore presents an avenue for prospective clinical modulation of inflammatory diseases as reviewed in ¹⁰⁵.

However, efferocytic macrophages might not be protective in all scenarios. In kidney, CD206+ macrophages with transcript expression characteristics of the efferocytic S1 macrophages have been shown to cause renal fibrosis.^{106,107} In line, the S1-promoting niche laminin- $\alpha 2$ was found to accumulate in the fibrotic kidney in a genetic model of Alport disease.¹⁰⁸ The hardening fibrotic renal tissue may further promote efferocytosis via Piezo-1 activation as well. Taken together, while both pro-inflammatory and pro-resolving mediators co-exist in the inflamed tissue, accumulative cell apoptosis acts as a timed switch to provoke efferocytosis which gradually tips the balance towards the resolving side. While efferocytosis modulation has immense therapeutic potential, caution must be taken in a pathology specific manner to evaluate its impacts on the disease.

11. Prospects in inflammatory tissue fate modulation via navigating macrophage cell states

The central role of macrophage efferocytosis in determining tissue inflammation fate and the beneficial effects presented by efferocytosis effectors in numerous pathology¹⁰⁴ emphasize vast therapeutic potential of macrophage cell state control and call for a holistic understanding. Nevertheless, this is a particularly challenging task consuming decades of research since macrophages present extreme functional and phenotypical diversity in vivo.

Since the last century, flexibility in macrophage response has been demonstrated in vitro: IFN- γ triggers a classical activation response,⁹⁶ and IL-4 triggers an alternative response marked by mannose receptor upregulation.⁹⁷ In 2000, the concept of M1/M2 first emerged to describe different in vivo macrophage tendencies using arginine to produce either cytotoxic nitric oxide (M1) or ornithine (M2) in different mouse strains.⁹⁸ Later studies associated M1 with the classical response, inflammation and cytotoxicity, and M2 with the alternative response and anti-inflammation. Analysing more in vitro stimulus-specific macrophage responses found the M1/M2 dichotomy over-simplistic, and M2 response was subdivided into M2a, M2b, M2c and M4.^{109,110} It has now been recognized that this conventional M1/M2 response scheme is insufficient to capture in vivo macrophage responses.¹¹¹ Transcriptome-wide analysis of tissue macrophages in bulk revealed macrophage states are flexibly shaped and thus distinct by tissue microenvironment.^{112,113}

Advances in transcriptomics at single-cell resolution provide further insights into the phenomenon of macrophages acquiring phenotypes specific to its residence tissue. Our recent meta-analysis of single-cell macrophage transcriptomes isolated from various healthy and pathological tissues available in public databases demonstrated that (a) tissue macrophages are heterogenous with both tissue-unique and tissue common cell states; (b) all tissue-common macrophages could be described by a vector of five gene fingerprint defined pan-tissue elemental identities (S1 to S5); (c) tissue monocytes show strong S3 identity and are more homogenous among tissues; and (d) proportion of macrophage identities differs by tissues.¹⁰⁰ These insights align with the emerging view that tissue macrophage is of heterogenous nature and explain the vast inter-tissue difference observed in bulk analysis of tissue macrophages.^{113,114} was a mixed result of different identity proportion among tissues and the presence of tissue-specific cell states. To provide a more comprehensive archive of tissue macrophage cell states, we have expanded this archive (MIKA) to include more tissues and pathological conditions (15 tissues, 20 conditions). Focusing on tissue-common macrophages, this expansion reveals S4 identity has limited cell state defining efficiency and existence of S1-associated cell states (S1-stray zone) that cannot be described by the five identity elements (**Figure 3A**). Therefore, we replaced the previous S4 element with two new elements to annotate these S1-strays. An unannotated S1-stray may represent ambiguous cell states since cells there express much fewer genes than others. (**Figure 3B**, Supplementary Data 1–4). Consistent with the previous version and other studies,^{68,115,116} MIKA also confirms well-known tissue-specific macrophages in lung, brain, cavities and liver (**Figure 3C**).

Efferocytic macrophages crucial to regulate inflammatory tissue fate have S1^{hi} identity. However, as mentioned earlier, the fact that this identity is enhanced in some pathological conditions such as renal fibrosis calls for disease-specific macrophage modulation. MIKA thus provides a holistic reference of the in vivo macrophage landscape for mapping healthy, pathological and drug-modulated cell states to navigate proper vectoral shift of identity towards the normal state. This

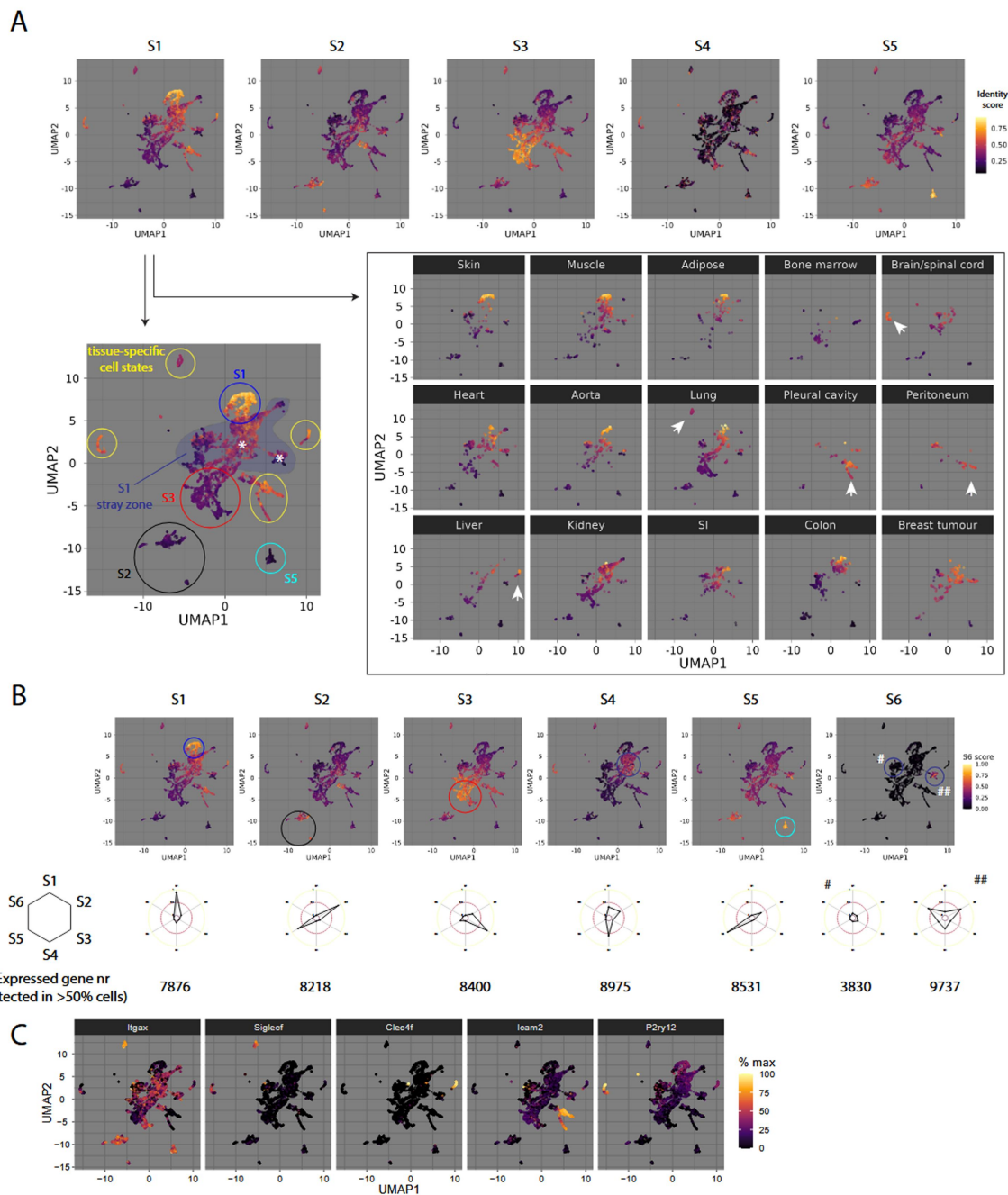


Figure 3

MIKA, the general gene signature-based identity framework for tissue macrophage.

MIKA currently covers 15 tissues and 20 conditions. (A) Distribution of the original five identity elements (S1-S5) were shown on UMAP. S1-associated stray zone (blue shade) and tissue-specific states (yellow circles and white arrows in tissue-expanded view) are evident. The original S4 identity is replaced by two surrogate identities (new S4 and S6) to S1 identity to annotate S1-strays (asterisks). (B) Radar maps show identity vectors of the indicated macrophage cell states. (C) Marker gene expression of tissue-specific macrophages (Itgax+Siglecf+ alveolar macrophage, Clec4f+ Kupffer cell, P2ry12+ microglia and Icam2+ cavity macrophages).

common reference also enables comparison of macrophage cell states described by different studies. Future quests to identify and associate macrophage cell state modulators (both known and new) to a particular identity shall facilitate this navigation for clinical benefits.

12. Concluding remarks

Extracellular matrix proteins are tightly coupled to the journey of a circulating leukocyte entering inflamed tissue. The vascular basement membrane anchors the endothelium to allow generation of cellular force required to remodel junctions for leukocyte passage. Leukocytes prefer using specialised diapedesis portals in the basement membrane, with weakened nearby endothelial junctions, to cross the vascular barriers. Optimal tissue entry requires aftercare of the “diapedesis pore” left behind by the extravasated leukocyte, that is fulfilled by cooperative efforts of collagens in basement membrane, platelets and EC. Both the vascular barrier passage and the myriad dynamic interaction with tissue cells and matrices influence leukocyte behaviours in tissue. These leukocyte activities, when properly orchestrated with macrophage fate, allow swift and timely restoration back to tissue homeostasis. This extravasation journey of leukocyte illustrates the intermingled relationship among extracellular matrices, leukocyte extravasation, tissue activities of extravasated leukocytes and the final tissue fate. We are still far from complete understanding of this extreme complexity. Yet, persistent stepwise advances in our understanding shall offer plenty regulatory targets for future modulation of inflammatory pathology.

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Additional information

Author contributions

Y.-T.L. wrote and edited the manuscript.

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Additional files

Supplementary data 1. Update documentation of MIKA. The text describes minor modification of identity score computation, addition of new identity elements auxiliary to S1 identity, and conditions and GEO accessions of newly added datasets. [↗](#)

Supplementary data 2. Macrophage identity scores of multiple tissue conditions in MIKA. Sample wise macrophage identity scores (complete or omitting monocyte-rich S3), tissue origin, perturbation, GEO accession and the definition of the input cell population for single-cell RNA sequencing were indicated. Note that the definition of the input population affects subsequent efficiency of *in silico* purification of

macrophages and thus the absolute identity scores in a dataset. This results in discrepancy of baseline identity between datasets of the same tissue (eg. GSE180420 and GSE200115). Identity kinetics is best assessed with the same definition of the input population (usually within the same dataset). [↗](#)

Supplementary data 3. Gene correlation to each macrophage identity in MIKA. For each tissue (baseline or perturbed), Pearson correlation coefficients of each gene to each of the macrophage identity (S1 to S6) are tabulated. Moran I coefficient shows how focused the gene expression is on UMAP. For searches of identity markers, the correlation should be considered in adjunct with gene expression; a gene of high correlation to an identity but with low expression likely underrepresents the identity of interest. [↗](#)

Supplementary data 4. Gene expression of each elemental macrophage identity in MIKA. For each tissue (baseline or perturbed), gene expression of each macrophage identity, represented by an identity score >0.5 or the top 10 scoring pseudobulks, is tabulated in natural logarithmic scale. Note that the matrix provides gross expression information for identity marker search but does not account for macrophages with compound identity as shown in Figure 3A. [↗](#)

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Reviewer #1 (Public review):

Summary:

In this review, the author covered several aspects of the inflammation response, mainly focusing on the mechanisms controlling leukocyte extravasation and inflammation resolution.

Strengths:

This review is based on an impressive number of sources, trying to comprehensively present a very broad and complex topic.

Weaknesses:

- (1) This reviewer feels that, despite the title, this review is quite broad and not centred on the role of the extracellular matrix.
- (2) The review will benefit from a stronger focus on the specific roles of matrix components and dynamics, with more informative subheadings.
- (3) The macrophage phenotype section doesn't seem well integrated with the rest of the review (and is not linked to the ECM).
- (4) Table 1 is difficult to follow. It could be reformatted to facilitate reading and understanding
- (5) Figure 2 appears very complex and broad.
- (6) Spelling and grammar should be thoroughly checked to improve the readability.

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Reviewer #2 (Public review):

Summary:

The manuscript is a timely and comprehensive review of how the extracellular matrix (ECM), particularly the vascular basement membrane, regulates leukocyte extravasation, migration, and downstream immune function. It integrates molecular, mechanical, and spatial aspects of ECM biology in the context of inflammation, drawing from recent advances. The framing of ECM as an active instructor of immune cell fate is a conceptual strength.

Strengths:

- (1) Comprehensive synthesis of ECM functions across leukocyte extravasation and post-transmigration activity.
- (2) Incorporation of recent high-impact findings alongside classical literature.
- (3) Conceptually novel framing of ECM as an active regulator of immune function.
- (4) Effective integration of molecular, mechanical, and spatial perspectives.

Weaknesses:

- (1) Insufficient narrative linkage between the vascular phase (Sections 2-6) and the in-tissue phase (Sections 7-10).
- (2) Underrepresentation of lymphocyte biology despite mention in early sections.
- (3) The MIKA macrophage identity framework is only loosely tied to ECM mechanisms.
- (4) Limited discussion of translational implications and therapeutic strategies.
- (5) Overly dense figure insets and underdeveloped links between ECM carryover and downstream immune phenotypes.
- (6) Acronyms and some mechanistic details may limit accessibility for a broader readership.

<https://doi.org/10.7554/eLife.108284.1.sa2>

Reviewer #3 (Public review):

Summary & Strengths:

This review by Yu-Tung Li sheds new light on the processes involved in leukocyte extravasation, with a focus on the interaction between leukocytes and the extracellular matrix. In doing so, it presents a fresh perspective on the topic of leukocyte extravasation, which has been extensively covered in numerous excellent reviews. Notably, the role of the extracellular matrix in leukocyte extravasation has received relatively little attention until recently, with a few exceptions, such as a study focusing on the central nervous system (J Inflamm 21, 53 (2024) doi.org/10.1186/s12950-024-00426-6) and another on transmigration hotspots (J Cell Sci (2025) 138 (11): jcs263862 doi.org/10.1242/jcs.263862). This review synthesizes the substantial knowledge accumulated over the past two decades in a novel and compelling manner.

The author dedicates two sections to discussing the relevant barriers, namely, endothelial cell-cell junctions and the basement membrane. The following three paragraphs address how leukocytes interact with and transmigrate through endothelial junctions, the mechanisms supporting extravasation, and how minimal plasma leakage is achieved during this process. The subsequent question of whether the extravasation process affects leukocyte differentiation and properties is original and thought-provoking, having received limited consideration thus far. The consequences of the interaction between leukocytes and the extracellular matrix, particularly regarding efferocytosis, macrophage polarization, and the outcome of inflammation, are explored in the subsequent three chapters. The review concludes by examining tissue-specific states of macrophage identity.

Weaknesses:

Firstly, the first ten sections provide a comprehensive overview of the topic, presenting logical and well-formulated arguments that are easily accessible to a general audience. In

stark contrast, the final section (Chapter 11) fails to connect coherently with the preceding review and is nearly incomprehensible without prior knowledge of the author's recent publication in *Cell. Mol. Life Sci.* CMLS 772 82, 14 (2024). This chapter requires significantly more background information for the general reader, including an introduction to the Macrophage Identity Kinetics Archive (MIKA), which is not even introduced in this review, its basis (meta-analysis of published scRNA-seq data), its significance (identification of major populations), and the reasons behind the revision of the proposed macrophage states and their further development. Secondly, while the attempt to integrate a vast amount of information into fewer figures is commendable, it results in figures that resemble a complex puzzle. The author may consider increasing the number of figures and providing additional, larger "zoom-in" panels, particularly for the topics of clot formation at transmigration hotspots and the interaction between ECM/ECM fragments and integrins. Specifically, the color coding (purple for leukocyte $\alpha 6$ -integrins, blue for interacting laminins, also blue for EC $\alpha 6$ integrins, and red for interacting 5-1-1 laminins) is confusing, and the structures are small and difficult to recognize.

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Author response:

Public Reviews:

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- (4) Table 1 is difficult to follow. It could be reformatted to facilitate reading and understanding*
- (5) Figure 2 appears very complex and broad.*
- (6) Spelling and grammar should be thoroughly checked to improve the readability.*

This review focuses on the whole extravasation journey of leukocyte and highlights involvement of extracellular matrix (ECM) in multiple phases of the process. ECM may exert their roles either as a collective structure or as individual components. In the revision, for those functions involving specific matrix components, we will emphasize the matrix

components and incorporate this information to subheadings as suggested. The parts of macrophage phenotype (Section 10-11) are included for its pivotal roles on deciding the tissue fate following inflammation (ie. to resolve / to regenerate damages incurred or to sustain inflammation), which is an important aspect of this review. ECM could modify macrophage phenotypes either directly (section 10) or indirectly via modulations of tissue stiffness or other cell types like fibroblasts (section 9). However, as pointed out by other reviewers as well, we acknowledge that Section 11 does not integrate well enough to the rest of the review. We plan to reorganize this part and to emphasize its link to ECM during the revision for better integration. We will reformat Table 1 for easier comprehension. We will consider restructuring Figure 2, which outlines various events influencing tissue decision of resolution/inflammation, perhaps by breaking up into two separate figures, to better focus the message. We will also check the language to improve readability.

Reviewer #2 (Public review):

Summary:

The manuscript is a timely and comprehensive review of how the extracellular matrix (ECM), particularly the vascular basement membrane, regulates leukocyte extravasation, migration, and downstream immune function. It integrates molecular, mechanical, and spatial aspects of ECM biology in the context of inflammation, drawing from recent advances. The framing of ECM as an active instructor of immune cell fate is a conceptual strength.

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- (5) Overly dense figure insets and underdeveloped links between ECM carryover and downstream immune phenotypes.*
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We will add a transition paragraph between Section 6 and Section 7 to provide a narrative that the extravasation processes affect downstream leukocyte functions. While lymphocytes follow a similar extravasation principle, their in-tissue activities differ from innate leukocytes. We will thus include discussion of lymphocyte-ECM crosstalk to Section 8 and/or 9 in the revision. We will restructure Section 11 and Figure 3 to better integrate to the rest of the review: In the current manuscript, we merely describe the capability of the MIKA

framework to describe identity of any tissue macrophages and thus the framework could serve as a roadmap to facilitate identity normalization of pathological macrophages. We plan, in the revision, by employing the MIKA framework, to discuss and demonstrate linkage between macrophage identities and expression/production of modulators to functional ECM effectors described in Section 8-9. Regarding the comment of limited discussion of translational implications / therapeutic strategies, we will try to enrich this aspect throughout the manuscript where appropriate, in addition to the existing ones (eg. line 293-297; 388-391; 460-463; 512-517) We will also revise figure structure in general to avoid too dense information and to improve clarity. We will consider to provide a glossary explaining specialized terms to expand readership accessibility.

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This review by Yu-Tung Li sheds new light on the processes involved in leukocyte extravasation, with a focus on the interaction between leukocytes and the extracellular matrix. In doing so, it presents a fresh perspective on the topic of leukocyte extravasation, which has been extensively covered in numerous excellent reviews. Notably, the role of the extracellular matrix in leukocyte extravasation has received relatively little attention until recently, with a few exceptions, such as a study focusing on the central nervous system (J Inflamm 21, 53 (2024) doi.org/10.1186/s12950-024-00426-6) and another on transmigration hotspots (J Cell Sci (2025) 138 (11): jcs263862 doi.org/10.1242/jcs.263862). This review synthesizes the substantial knowledge accumulated over the past two decades in a novel and compelling manner.

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interacting 5-1-1 laminins) is confusing, and the structures are small and difficult to recognize.

We agree with and appreciate the specific and helpful suggestions by the reviewer. During the revision, we will provide the requested background description of MIKA to enhance accessibility of general readership. As pointed out by other reviewers, since this part (Section 11) is less well-integrated to the rest of the review, we will restructure this part by linking tissue macrophage identities under MIKA framework to modulation of functional ECM effectors described in previous sections (Section 8-9). We acknowledge the current figure organization might be overly information-dense and will consider breaking down the contents to multiple figures. The size and color-coding issues will also be addressed.

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